

Question (1): (25 Marks)

- a- Sketch an architecture diagram of Faster R-CNN for object detection. Explain why it is called a two-stage object detector. (15 Marks)
- b- Consider a task of image classification where you want to classify each image with a label and each image contains only one object. The feature extraction step will be handcrafted where the extracted features will be passed to some classifier. Do you think that it should be better to use SIFT or HOG as the feature extraction algorithm for this task? Explain your choice. (10 Marks)

Question (2): (25 Marks)

- a- Given the following 2-D point cloud, it is supposed to fit an optimal line model to it. (12 Marks)



(a) Original point cloud



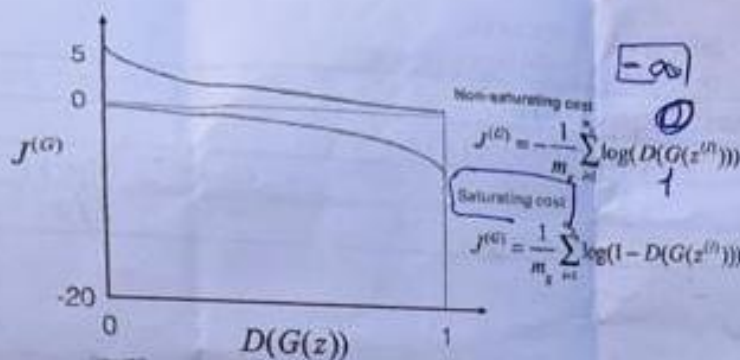
(b) Line model after fitting

- i. Describe how you will use **RANSAC** algorithm for the aforementioned task.
- ii. Assume that there are 65% inliers; compute the minimal number of RANSAC iterations needed to get, with probability 90%, at least one random sample that is free from outliers.
- b- In the **Harris** feature detector (13 Marks)
- i. Describe what is the second-moment matrix? Explain its role and how this is used to select good corners.
- ii. Given the following second-moment matrix $M = \begin{bmatrix} 403 & 385 \\ 385 & 381 \end{bmatrix}$, compute the Harris response function R with $\alpha = 0.04$. Do you think that this value indicates the existence of a corner, edge or a flat area?

Question (3): (25 Marks)

$\leftarrow \frac{|m|}{n} = \alpha \text{ (Trace)}$

- a- Consider the graph in the given figure representing the training procedure of a GAN. (10 Marks)
- i. Early in the training, is the value of $D(G(z))$ closer to 0 or closer to 1? Explain.
- ii. Two cost functions are presented in the figure, which one would you choose to train your GAN? Justify your answer. Cost 1
- iii. You know that your GAN is trained when $D(G(z))$ is close to 1. ☒ True ☐ False? Explain.



- b- Consider in a given task, that you have a pair of images containing multiple objects that have undergone motion from one frame to the next. Your objective is to segment the image sequence into a finite number of "motion layers." A motion layer is a subset of pixels in the pair of frames that move together according to a given motion model. **Explain** a complete algorithm that takes that pair of images and produces the set of motion layers. Consider using the optical flow which you have learnt in class as your motion model to calculate the motion. (15 **Marks**)

Question (4): (25 Marks)

- a- Given a video sequence of a city traffic with moving pedestrians, a Multiple Object Tracking (MOT) system can be designed to track all pedestrians in the video. The tracker can get benefit from a pedestrians' detector, where the detector is responsible of detecting pedestrians in each frame, and the tracker is responsible of associating detections from time frame t to frame $t+1$.

Define what kind of error(s) this tracker should handle for each of the following scenarios in the video sequence: (12 **Marks**)

1. Many partial/complete occlusions of pedestrians occur.
2. Many pedestrians cross each other.

- b- In the Siamese network for object verification, describe the triplet loss and what it is used for. Sketch a general architecture of the Network (13 **Marks**)